Base Cases and Induction Hypotheses

For the following claims, demonstrate the of the base cases, and write out the statement of the induction hypothesis. Then if you have time attempt to demonstrate the induction hypothesis step.

Claim 1

$2^0 + 2^1 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$ for $n \geq 0$, n an integer.

Claim 2

$1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n+1)! - 1$ for $n \geq 1$, n an integer.

Claim 3

$8^n - 3^n$ is divisible by 5 for all integers $n \geq 1$. Hint: Another way to say "divisible by 5" is $8^n - 3^n = 5k$ for some integer k.

Claim 4

$\sum_{i=1}^{n}(8i - 5) = 4n^2 - n$